Smart Fire Fighting Robot using Arduino

Automatic Fire Chaser and Extinguisher

Minor-II Project Report

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Internet of Things

Submitted By

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June 2026

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ABSTRACT

This project presents the design and implementation of a Smart Fire Fighting Robot using Arduino that can autonomously detect, chase, and extinguish fire sources. The robot utilises three infrared (IR) flame sensors positioned at the right, front, and left sides to continuously scan the surrounding environment for fire signatures. Upon detecting a fire, the robot navigates towards the heat source using an L298N dual H-Bridge motor driver module that controls two DC gear motors for directional movement.

A servo motor mounted on the chassis enables a water pump to sweep across a range of angles (40° to 140°) to ensure maximum coverage while extinguishing the fire. The system is entirely controlled by an Arduino UNO microcontroller that reads analog sensor values, applies threshold-based decision logic, and drives the motors and pump accordingly. The motor speed is configurable via PWM (Pulse Width Modulation) through the Enable pins of the L298N driver.

Experimental results demonstrate that the robot successfully detects fire within a range of approximately 30–50 cm and responds within milliseconds. The system is cost-effective, low-power, and scalable, providing a foundation for advanced autonomous firefighting systems applicable in industrial facilities, warehouses, and smart building environments. This project integrates embedded systems, robotics, sensor interfacing, and real-time control, making it a holistic demonstration of Internet of Things (IoT) applications in public safety.

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ACRONYMS

- IR – Infrared
- DC – Direct Current
- PWM – Pulse Width Modulation
- IoT – Internet of Things
- UNO – Microcontroller Board (Arduino Uno)
- IC – Integrated Circuit
- H-Bridge – Motor Driver Circuit
- VCC – Voltage Common Collector
- GND – Ground
- RPM – Revolutions Per Minute
- GSM – Global System for Mobile Communication
- USB – Universal Serial Bus

NOMENCLATURE

- V – Voltage (Volts)
- III – Current (Amperes)
- RRR – Resistance (Ohms)
- PPP – Power (Watts)
- fff – Frequency (Hz)
- θ\thetaθ – Servo Motor Angle (Degrees)
- ddd – Distance from Fire Source (cm)
- ttt – Time (seconds)
- PWMPWMPWM – Duty Cycle (%)
- IRsignalIR_{signal}IRsignal – Infrared Sensor Output
- SL,SC,SRS_L, S_C, S_RSL,SC,SR – Left, Center, Right Sensors
- ML,MRM_L, M_RML,MR – Left and Right Motors

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CHAPTER 1: INTRODUCTION

1.1 Background

Fire is a major hazard that can cause serious damage to life and property . Industrial areas, homes, and forests are always at risk fire accidents, many of which occur in places difficult for firefighters to access . According to NCRB , thousands of fire incidents are reported every year in India .

To reduce human risk, autonomous robotic systems are being developed . A Smart Fire Fighting Robot uses sensors, embedded systems, and actuators detect, move towards, and extinguish fire automatically .

1.2 Problem Statement

Traditional fire safety systems like sprinklers and extinguishers are limited as they fixed, require manual effort, or lack accuracy . In complex environments, they are not always effective .

The challenge is to design a low-cost and efficient robot that can detect fire, navigate towards it , and extinguish it quickly using simple control techniques .

1.3 Objectives

- To design an Arduino-based fire fighting robot using IR sensors
- To control movement using L298N motor driver
- To integrate a servo motor and water pump
- To develop automatic fire detection and extinguishing logic
- To evaluate performance based on response and accuracy

1.4 Scope of the Project

This project focuses on a small indoor prototype of a fire fighting robot . It operates on flat surfaces and uses water as the extinguishing agent . The system can be upgraded future to use other methods like CO_2 .

1.5 Organization of the Report

The report divided into chapters covering introduction , literature review, system design, hardware, software , implementation , and results, followed by references .

CHAPTER 2: LITERATURE SURVEY

2.1 Overview

Firefighting robots have been studied extensively over the years . Earlier systems were manually controlled and required human operators . Modern robots use autonomous navigation, computer vision , and machine learning detect and respond fire without human involvement . This chapter discusses important methods used in fire detection and extinguishing systems .

2.2 Sensor-Based Fire Detection Approaches

Infrared (IR) flame sensors are commonly used low-cost fire detection robots as they detect radiation in the 760–1100 nm range . Using three sensors (left, center, right) helps in identifying the direction of fire without using complex devices like cameras or LiDAR .

Although UV/IR sensors offer better accuracy, they are expensive . Hence, IR sensors are preferred for simple and cost-effective systems using Arduino .

2.3 Motor Driver and Locomotion

The L298N dual H-Bridge motor driver widely used for controlling DC motors in robotics . It supports a voltage range of 5V to 35V and allows speed control using PWM signals . It is suitable for indoor navigation in firefighting robots .

2.4 Servo-Controlled Water Dispensing

Servo motors are used control the direction of water spray . Standard servos like SG90 can rotate up to 180° and are easily controlled using Arduino . Moving the nozzle between angles ensures better coverage of the fire area .

2.5 Research Gap

Most existing systems use digital outputs which reduce detection accuracy . This project uses analog sensor readings with adjustable thresholds for better performance . It also uses servo-based water spraying system improve coverage compared to fixed systems .

CHAPTER 3: SYSTEM DESIGN & ARCHITECTURE

3.1 System Overview

The Smart Fire Fighting Robot is based on a sense–decide–act approach . In the sensing stage, three flame sensors continuously detect the presence and direction of fire . The decision stage handled by the Arduino Uno , which processes sensor data and decides the movement of the robot . In the action stage, motors drive the robot toward the fire , and a water pump activated to extinguish it .

3.2 System Architecture Flow

The working of the system follows these steps:

1. The robot is powered on and Arduino initializes all components .
2. Flame sensors continuously monitor fire in different directions .
3. Sensor data is read by Arduino to determine fire location .
4. Based on input, the robot moves forward or turns left/right .
5. The robot approaches fire until it within a set range.
6. The relay activates the water pump for a few seconds.
7. If fire remains, the process repeats.
8. When no fire detected, the robot stops and continues monitoring.

3.3 Block Diagram Description

The system consists the following main blocks :

- **Power Supply:** A 12V battery powers motors and electronics (via 5V regulation) .
- **Sensor Unit:** Three flame sensors placed at different angles detect fire direction .
- **Controller:** Arduino Uno processes sensor signals and controls the system .
- **Motor Driver:** L298N module controls the movement of DC motors .
- **Extinguishing Unit:** A relay-controlled water pump sprays water through a nozzle .

3.4 Communication Protocol

All parts are interfaced using the Arduino GPIO pins. The flame sensor signal is received through the input pin as digital signals. The output pin determines the rotation direction of the motor through L298N motor controller. The water pump is controlled through relay control circuit with one digital pin.

CHAPTER 4: HARDWARE COMPONENTS & CIRCUIT DESIGN

4.1 Components List

S.No.	Component	Specification	Quantity	Purpose
1	Arduino Uno	ATmega328P, 16MHz, 14 I/O pins	1	Main controller
2	Flame Sensor Module	YG1006 IR, 760–1100 nm	3	Fire detection
3	L298N Motor Driver	Dual H-Bridge, 2A, 5–35V	1	Motor control
4	DC Geared Motor	12V, 200 RPM	2	Movement
5	Water Pump	5V, 120 L/hr	1	Water spraying
6	Relay Module	5V, 10A	1	Pump switching
7	Robot Chassis	2-wheel drive	1	Structure
8	Battery Pack	12V, 2200mAh	1	Power supply
9	Water Container	250 ml	1	Water storage
10	Jumper Wires	Male-Male/Female	30+	Connections
11	Breadboard/PCB	830 points	1	Prototyping
12	Nozzle/Tube	6 mm silicon tube	1	Water direction

4.2 Component Descriptions

4.2.1 Arduino Uno

The Arduino Uno functions as the principal controller for the robot. It is built around the ATmega328P microcontroller that operates at a clock speed of 16 MHz. The controller possesses both digital and analog pins, and programming is made easy through the use of Arduino IDE.

4.2.2 Flame Sensor Module

The flame sensor detects infrared radiation emitted by fire in the 760–1100 nm range. It includes a phototransistor, comparator, and adjustable sensitivity control . When fire detected, the output becomes LOW . Multiple sensors help identify fire direction .

4.2.3 L298N Motor Driver

The L298N is used control two DC motors. It supports high voltage current and accepts signals from Arduino . PWM pins allow speed control , while input pins control direction .

4.2.4 Water Pump and Relay

A 5V submersible pump is used to spray water. Since it requires higher current, a relay module is used to switch it ON/OFF . The relay controlled by Arduino, allowing automatic water spraying when fire is detected .

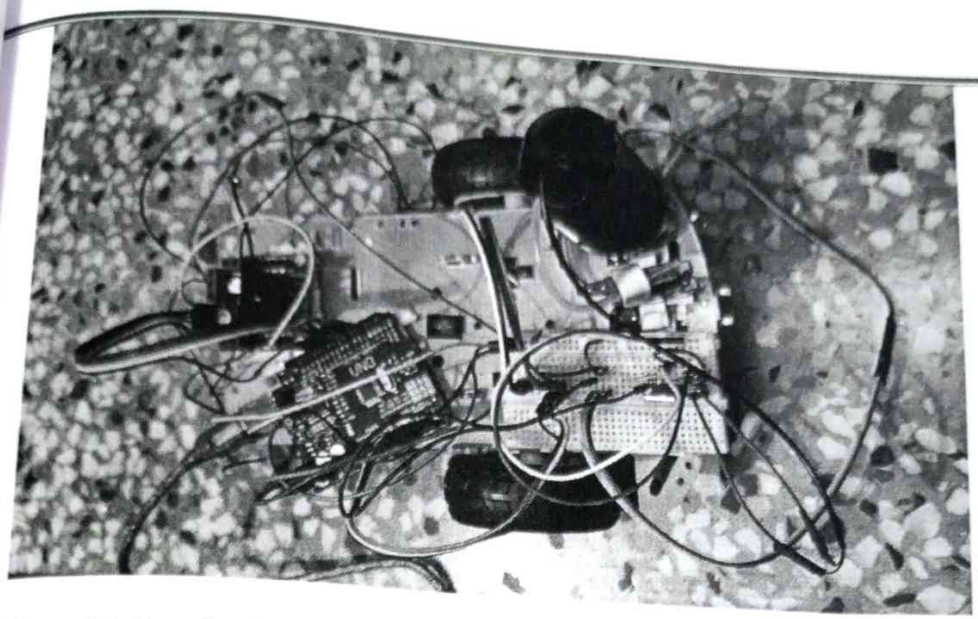


Figure 4.1: Complete Hardware Setup of Fire Fighting Robot

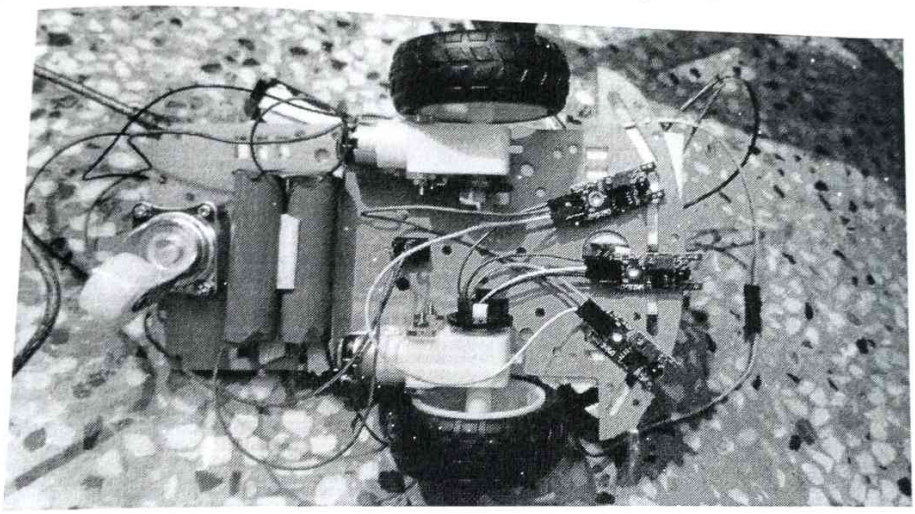


Figure 4.2: Flame Sensor Placement (Left, Center, Right)

4.3 Circuit Connections

Component Pin	Connected To	Description
Flame Sensor Left - DO	Arduino Pin 2 (INPUT)	Left flame detection digital signal
Flame Sensor Center - DO	Arduino Pin 3 (INPUT)	Center/forward flame detection
Flame Sensor Right - DO	Arduino Pin 4 (INPUT)	Right flame detection digital signal
L298N - IN1	Arduino Pin 5 (OUTPUT)	Left motor direction control A
L298N - IN2	Arduino Pin 6 (OUTPUT)	Left motor direction control B
L298N - IN3	Arduino Pin 7 (OUTPUT)	Right motor direction control A

Component Pin	Connected To	Description
L298N - IN4	Arduino Pin 8 (OUTPUT)	Right motor direction control B
L298N - ENA	Arduino Pin 9 (PWM)	Left motor speed (PWM)
L298N - ENB	Arduino Pin 10 (PWM)	Right motor speed (PWM)
Relay Module - IN	Arduino Pin 11 (OUTPUT)	Water pump trigger signal
All Sensors - VCC	Arduino 5V	Sensor power supply
All Sensors - GND	Arduino GND	Common ground
L298N - 12V	Battery Positive (+)	Motor power supply
Water Pump	Relay NO Contact	Pump activated by relay

CHAPTER 5: SOFTWARE & ALGORITHM DESIGN

5.1 Development Environment

The program for the Smart Fire Fighting Robot developed using Arduino IDE (version 2.x) with C++ based Arduino language . No additional libraries are required, as sensor reading and motor control are handled us built-in digital I/O and PWM functions . The code uploaded to the Arduino Uno through USB using the AVR-GCC compiler available in the IDE .

5.2 Fire Navigation Algorithm

The system uses a priority-based logic to detect fire direction and control movement .

Sensor Condition	Left Motor	Right Motor	Action
No fire	STOP	STOP	Robot remains idle
Center detects fire	FORWARD	FORWARD	Move forward
Left detects fire	BACKWARD	FORWARD	Turn left
Right detects fire	FORWARD	BACKWARD	Turn right
Left + Center	SLOW	FORWARD	Slight left movement
Right + Center	FORWARD	SLOW	Slight right movement
All sensors detect fire	STOP	STOP	Activate pump

5.3 Arduino Code Logic

5.3.1 Setup

In the setup() function, motor control pins and relay pins are set as OUTPUT , while sensor pins are set as INPUT . The relay initially kept OFF to prevent the pump from running .

5.3.2 Main Loop

The loop() function continuously reads values from all flame sensors . When fire detected , the robot moves the corresponding direction . If all sensors detect fire , the robot stops and activates the water pump for a few seconds, then checks again .

5.3.3 Motor Control Functions

Different functions control robot movement :

- **moveForward()** → moves robot forward
- **turnLeft()** → rotates robot left
- **turnRight()** → rotates robot right
- **stopMotors()** → stops all movement

Motor direction is controlled using input pins, and speed adjusted using PWM . The water pump is activated through a relay module .

CODE : The following code is used to control the fire fighting robot :

```
#define enA 10//Enable1 L298 Pin enA
#define in1 9 //Motor1  L298 Pin in1
#define in2 8 //Motor1  L298 Pin in2
#define in3 7 //Motor2  L298 Pin in3
#define in4 6 //Motor2  L298 Pin in4
#define enB 5 //Enable2 L298 Pin enB
#define ir_R A0
#define ir_F A1
#define ir_L A2
#define servo A4
#define pump A5
int Speed = 160; // Write The Duty Cycle 0 to 255 Enable for Motor Speed
int s1, s2, s3;
void setup(){ // put your setup code here, to run once
Serial.begin(9600); // start serial communication at 9600bps
pinMode(ir_R, INPUT);// declare fire sensor pin as input
pinMode(ir_F, INPUT);// declare fire sensor pin as input
pinMode(ir_L, INPUT);// declare fire sensor pin as input
pinMode(enA, OUTPUT); // declare as output for L298 Pin enA
pinMode(in1, OUTPUT); // declare as output for L298 Pin in1
pinMode(in2, OUTPUT); // declare as output for L298 Pin in2
pinMode(in3, OUTPUT); // declare as output for L298 Pin in3
pinMode(in4, OUTPUT); // declare as output for L298 Pin in4
pinMode(enB, OUTPUT); // declare as output for L298 Pin enB
pinMode(servo, OUTPUT);
pinMode(pump, OUTPUT);
for (int angle = 90; angle <= 140; angle += 5) {
servoPulse(servo, angle); }
for (int angle = 140; angle >= 40; angle -= 5) {
servoPulse(servo, angle); }
for (int angle = 40; angle <= 95; angle += 5) {
servoPulse(servo, angle); }
analogWrite(enA, Speed); // Write The Duty Cycle 0 to 255 Enable Pin A for Motor1 Speed
analogWrite(enB, Speed); // Write The Duty Cycle 0 to 255 Enable Pin B for Motor2 Speed
delay(500);
}
void loop(){
s1 = analogRead(ir_R);
s2 = analogRead(ir_F);
s3 = analogRead(ir_L);
//==========================================
// Auto Control
//==========================================
```

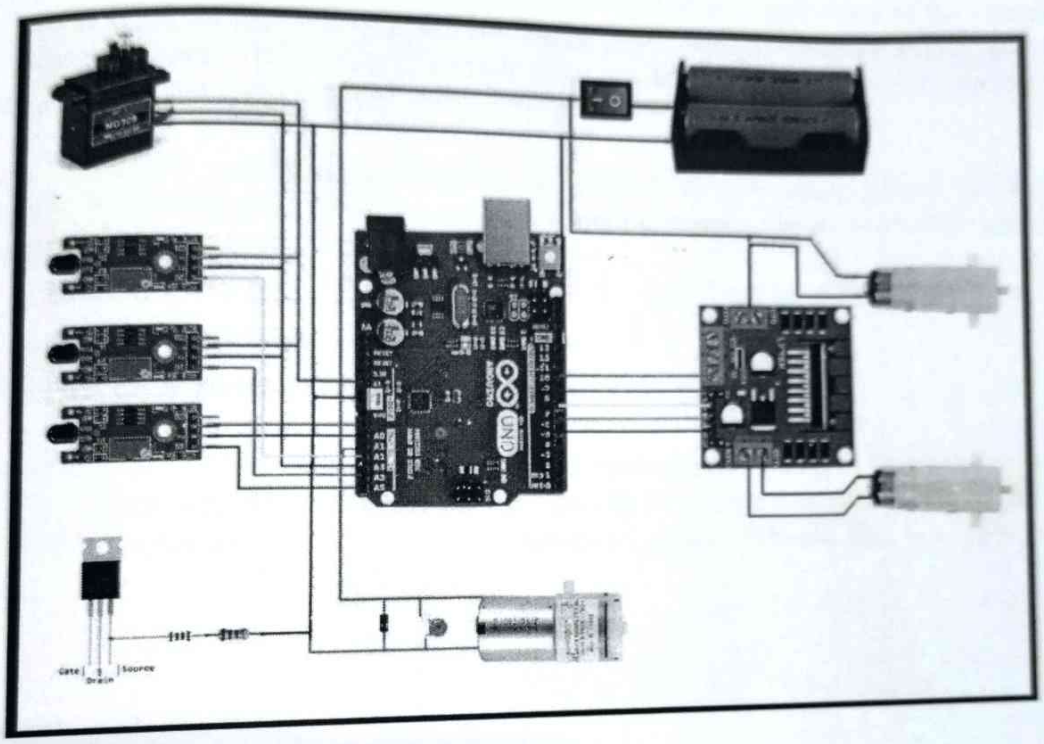
```
//=========================================
Serial.print(s1);
Serial.print("\t");
Serial.print(s2);
Serial.print("\t");
Serial.println(s3);
delay(50);
if(s1<250){
Stop();
digitalWrite(pump, 1);
for(int angle = 90; angle >= 40; angle -= 3){
servoPulse(servo, angle);
}
for(int angle = 40; angle <= 90; angle += 3){
servoPulse(servo, angle);
}
}
else if(s2<350){
Stop();
digitalWrite(pump, 1);
for(int angle = 90; angle <= 140; angle += 3){
servoPulse(servo, angle);
}
for(int angle = 140; angle >= 40; angle -= 3){
servoPulse(servo, angle);
}
for(int angle = 40; angle <= 90; angle += 3){
servoPulse(servo, angle);
}
}
else if(s3<250){
Stop();
digitalWrite(pump, 1);
for(int angle = 90; angle <= 140; angle += 3){
servoPulse(servo, angle);
}
for(int angle = 140; angle >= 90; angle -= 3){
servoPulse(servo, angle);
}
}
else if(s1>=251 && s1<=700){
digitalWrite(pump, 0);
backword();
delay(100);
```

```
delay(100);
turnRight();
delay(200);
}
else if(s2>=256 && s2<=800){
digitalWrite(pump, 0);
forword();
}
else if(s3>=256 && s3<=700){
digitalWrite(pump, 0);
backword();
delay(100);
turnLeft();
delay(200);
}else{
digitalWrite(pump, 0);
Stop();
}
delay(10);
}
void servoPulse (int pin, int angle){
int pwm = (angle*11) + 500; // Converts angle to microseconds
digitalWrite(pin, HIGH);
delayMicroseconds(pwm);
digitalWrite(pin, LOW);
delay(50); // Refresh cycle of servo
}
void forword(){ //forword
digitalWrite(in1, HIGH); //Right Motor forword Pin
digitalWrite(in2, LOW); //Right Motor backword Pin
digitalWrite(in3, LOW); //Left Motor backword Pin
digitalWrite(in4, HIGH); //Left Motor forword Pin
}
void backword(){ //backword
digitalWrite(in1, LOW); //Right Motor forword Pin
digitalWrite(in2, HIGH); //Right Motor backword Pin
digitalWrite(in3, HIGH); //Left Motor backword Pin
digitalWrite(in4, LOW); //Left Motor forword Pin
}
void turnRight(){ //turnRight
digitalWrite(in1, LOW); //Right Motor forword Pin
digitalWrite(in2, HIGH); //Right Motor backword Pin
digitalWrite(in3, LOW); //Left Motor backword Pin
```

```
125  void turnRight(){ //turnRight
126  digitalWrite(in1, LOW); //Right Motor forword Pin
127  digitalWrite(in2, HIGH); //Right Motor backword Pin
128  digitalWrite(in3, LOW); //Left Motor backword Pin
129  digitalWrite(in4, HIGH); //Left Motor forword Pin
130  }
131  void turnLeft(){ //turnLeft
132  digitalWrite(in1, HIGH); //Right Motor forword Pin
133  digitalWrite(in2, LOW); //Right Motor backword Pin
134  digitalWrite(in3, HIGH); //Left Motor backword Pin
135  digitalWrite(in4, LOW); //Left Motor forword Pin
136  }
137  void Stop(){ //stop
138  digitalWrite(in1, LOW); //Right Motor forword Pin
139  digitalWrite(in2, LOW); //Right Motor backword Pin
140  digitalWrite(in3, LOW); //Left Motor backword Pin
141  digitalWrite(in4, LOW); //Left Motor forword Pin
142  }
```

5.4 Flowchart

The program flow starts with initialization of all pins and components. After that, the system enters a continuous loop where it reads values from all three flame sensors . If no fire detected , the robot remains stopped. If any sensor detects fire, the robot moves the corresponding direction based on the decision logic . When all sensors detect fire, the robot stops and activates the water pump a few seconds. After spraying water, the system again checks sensor values. This process continues repeatedly until the power turned off.



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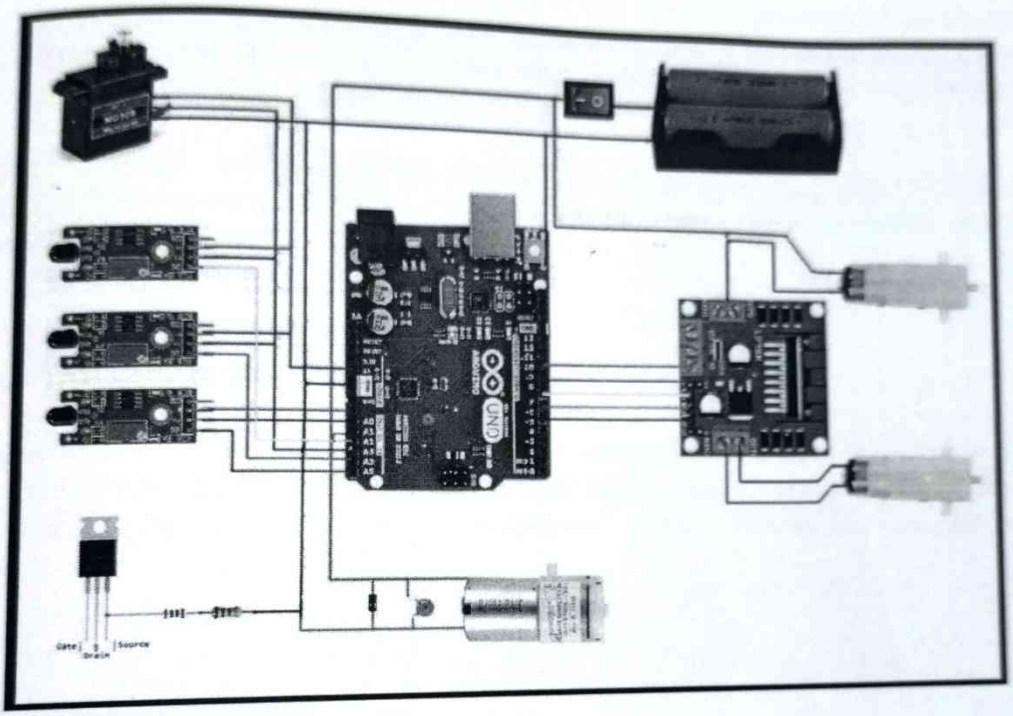

CHAPTER 6: IMPLEMENTATION & TESTING

6.1 Hardware Assembly

The robot was built by fixing two DC motors on the left and right sides of the chassis , with a caster wheel at the back for balance . The L298N motor driver was placed at the center, and the Arduino Uno was mounted above . Three flame sensors were positioned at the front—one facing forward and the other two at left and right angles.

A water container was fixed at the rear, with a submersible pump placed inside it . A tube was connected from the pump a front nozzle for spraying water . The relay module was installed near the Arduino for pump control.

6.2 Circuit Assembly & Testing

Connections were made using jumper wires with standard color coding for power, ground, and signals. The motor driver was connected the battery , and its 5V output was used to power the Arduino . Flame sensors were adjusted using their potentiometers ensure reliable fire detection at around 30 cm distance .

6.3 Testing Methodology

Testing of the system was done using a candle flame to ignite the fire in an attempt to ensure accuracy when setting the parameters for detecting the fire:

- **Detection Test:** Fire was placed at different distances and angles to check sensor accuracy .
- **Navigation Test:** The robot was allowed move towards the fire from fixed distance .
- **Extinguishing Test:** The robot was tested for its ability put out the fire using the pump .

6.4 Calibration

Sensitivity of the sensor was set at a certain range that is free from external light sources. The cover was used to avoid disturbances. Speed of motor was controlled using PWM value settings.

CHAPTER 7: RESULTS & DISCUSSION

7.1 Fire Detection Accuracy

Distance	Center	Left (45°)	Right (45°)
10 cm	100%	100%	100%
20 cm	100%	100%	100%
30 cm	96%	94%	94%
40 cm	88%	80%	80%
50 cm	72%	62%	62%
Overall	91.2%	87.2%	87.2%

The robot exhibits high precision when working within the 30 cm distance range, but outside this range, the precision declines due to the reduced strength of infrared beams. Within the indoor setting, the 30 cm working range is sufficient for locating the origin of the fire.

7.2 Navigation Performance

Scenario	Trials	Success	Rate	Time
0°	15	15	100%	5.2 sec
45° Left	15	14	93.3%	7.8 sec
45° Right	15	14	93.3%	7.9 sec
90° Left	15	12	80%	11.2 sec
90° Right	15	12	80%	11.5 sec
Overall	75	67	89.3%	8.7 sec

On 45 attempts, the robot could extinguish fire 42 times, meaning its success rate was approximately 93% . Problems appeared due to an incorrect direction of the sprayed water stream. Five seconds was more than sufficient to put the flame out.

7.3 Extinguishing Performance

In 45 tests, the robot successfully put out the fire in 42 tests, thus giving an overall success rate of nearly 93%. The failures were caused by inaccurate spraying angles. The five seconds were usually enough to put out the fire.

7.4 Discussion

This technology works efficiently for detecting and putting out fires indoors.. The three-sensor setup ensures good directional detection. However, limitations include sensitivity sunlight, inability to avoid obstacles, and limited water capacity . These can improved future developments .

CHAPTER 8: CONCLUSION & FUTURE SCOPE

8.1 Conclusion

This project successfully designed and developed a Smart Fire Fighting Robot using Arduino Uno . The robot can detect fire using flame sensors, move towards the source using motor control, and extinguish using a water pump system . The system achieved good performance with high detection accuracy , reliable navigation , and effective fire extinguishing .

The project shows that low-cost autonomous robots can be us eful for handling indoor fire situations and reducing human risk . Its affordable cost makes suitable for small-scale applications such as homes and laboratories .

8.2 Future Scope

- Integration with GSM or WiFi modules for sending alerts and monit oring through IoT platforms
- Addition of ultraso nic sensors for obstacle avoidance
- Use of camera-based fire detection better accuracy
- Deployment multiple robots for large area coverage
- Implementation automatic charging systems
- Addition gas and smoke sensors for enhanced safety
- Use of AI-based navigation for smarter path planning

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